

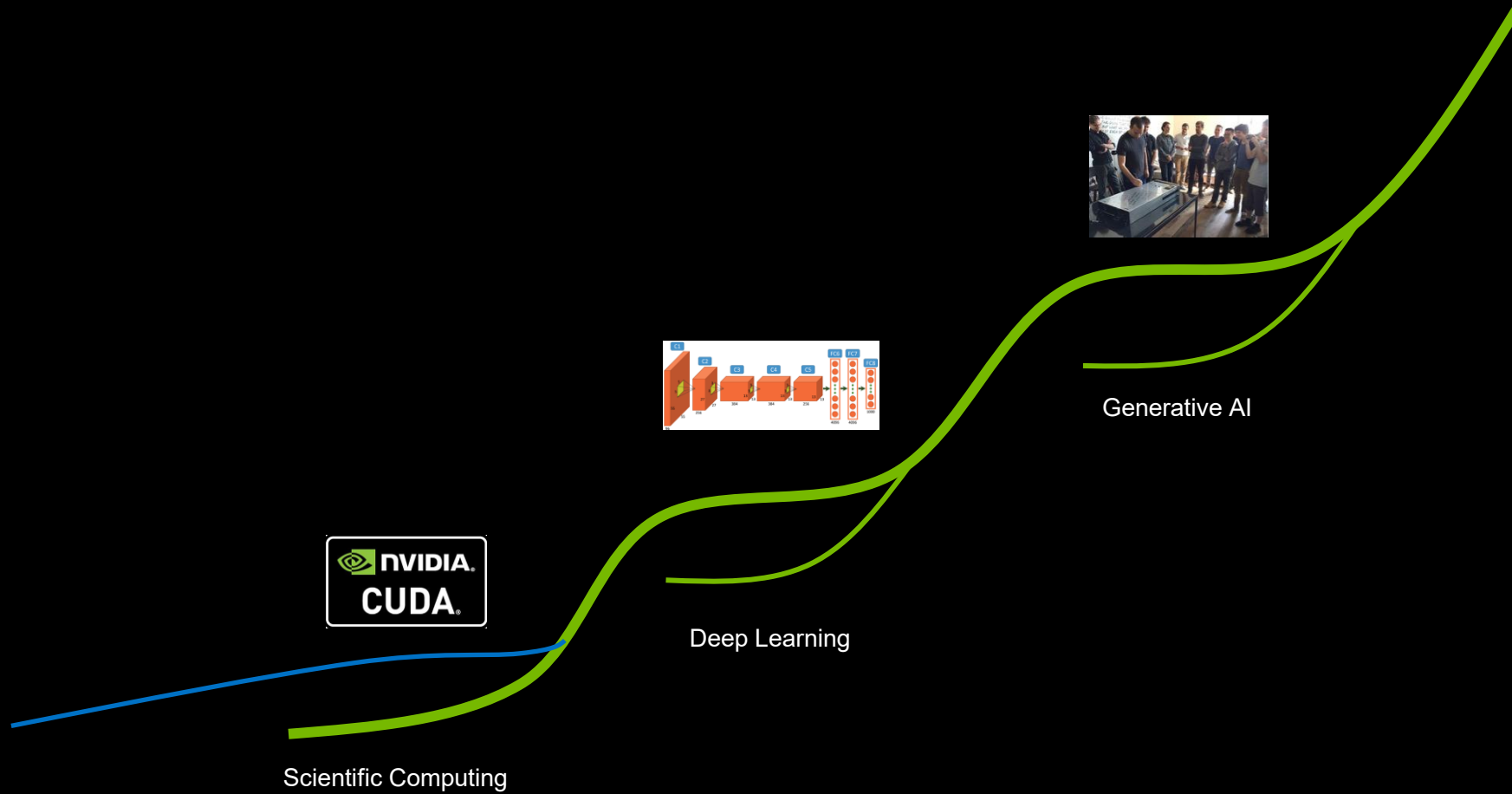
Scalable Validation and Optimized Simulation for Quantum Machine Learning

Tutorials · QCE 26

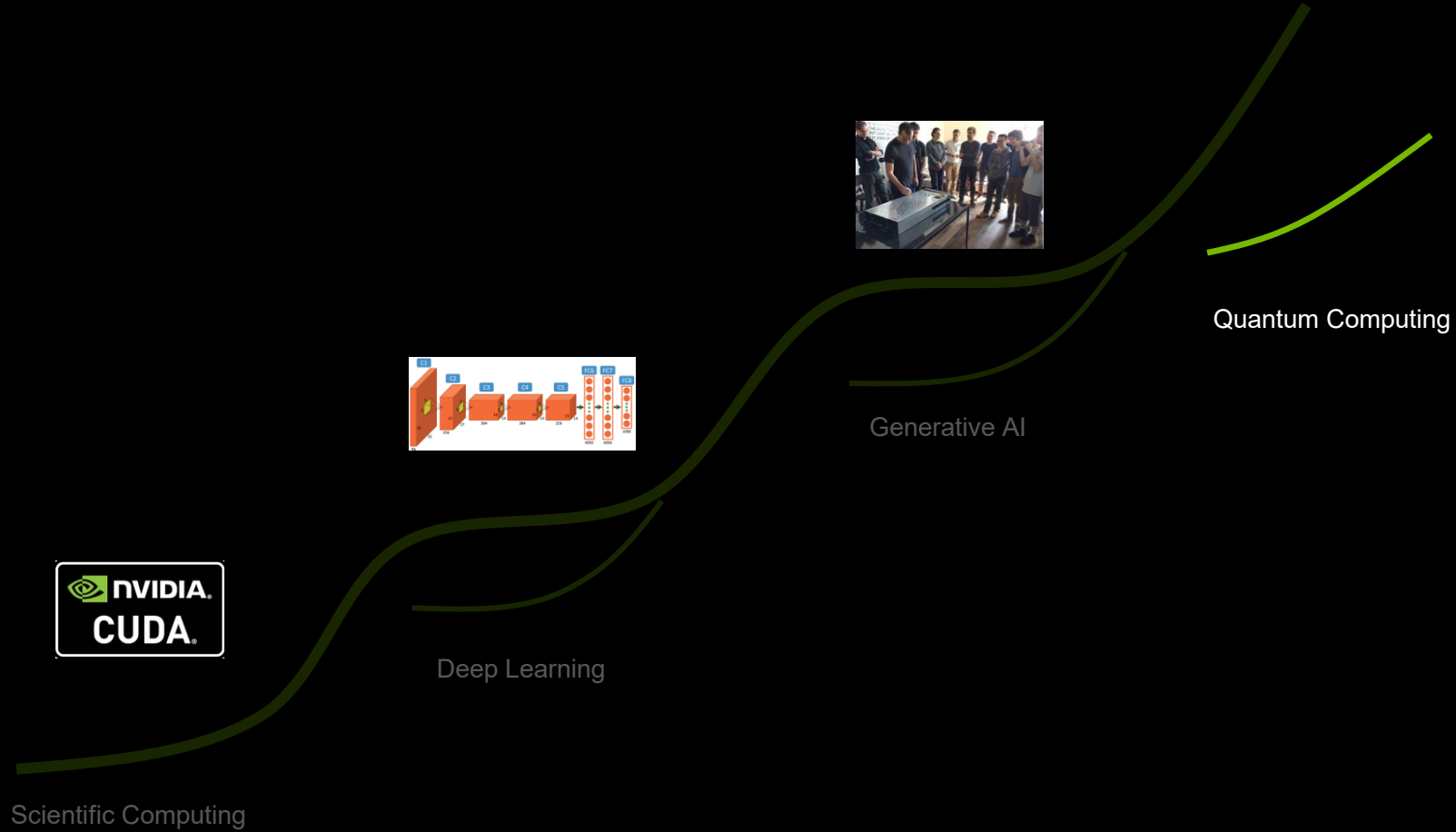
Yun Yuan (Pika) Wang, Sr. Quantum Solutions Architect | NVIDIA
Kuan-Cheng (Louis) Chen, Lead Quantum Researcher | Imperial College London / Jij Inc.
Tai-Yue Li, Sr. Researcher | NCHC

Jiun-Cheng Jiang, Solution Architect | NVIDIA
Kevin Ho, Researcher | NCHC

NVIDIA's History of Enabling Computing Revolutions



NVIDIA's History of Enabling Computing Revolutions





NVIDIA is not building
Qubits



NVIDIA is building all
Accelerated Quantum
Supercomputers

AI and Quantum

AI to Advance Quantum Computing



Algorithms

QEC

Calibration

Digital Twins

Quantum Computing to Advance AI



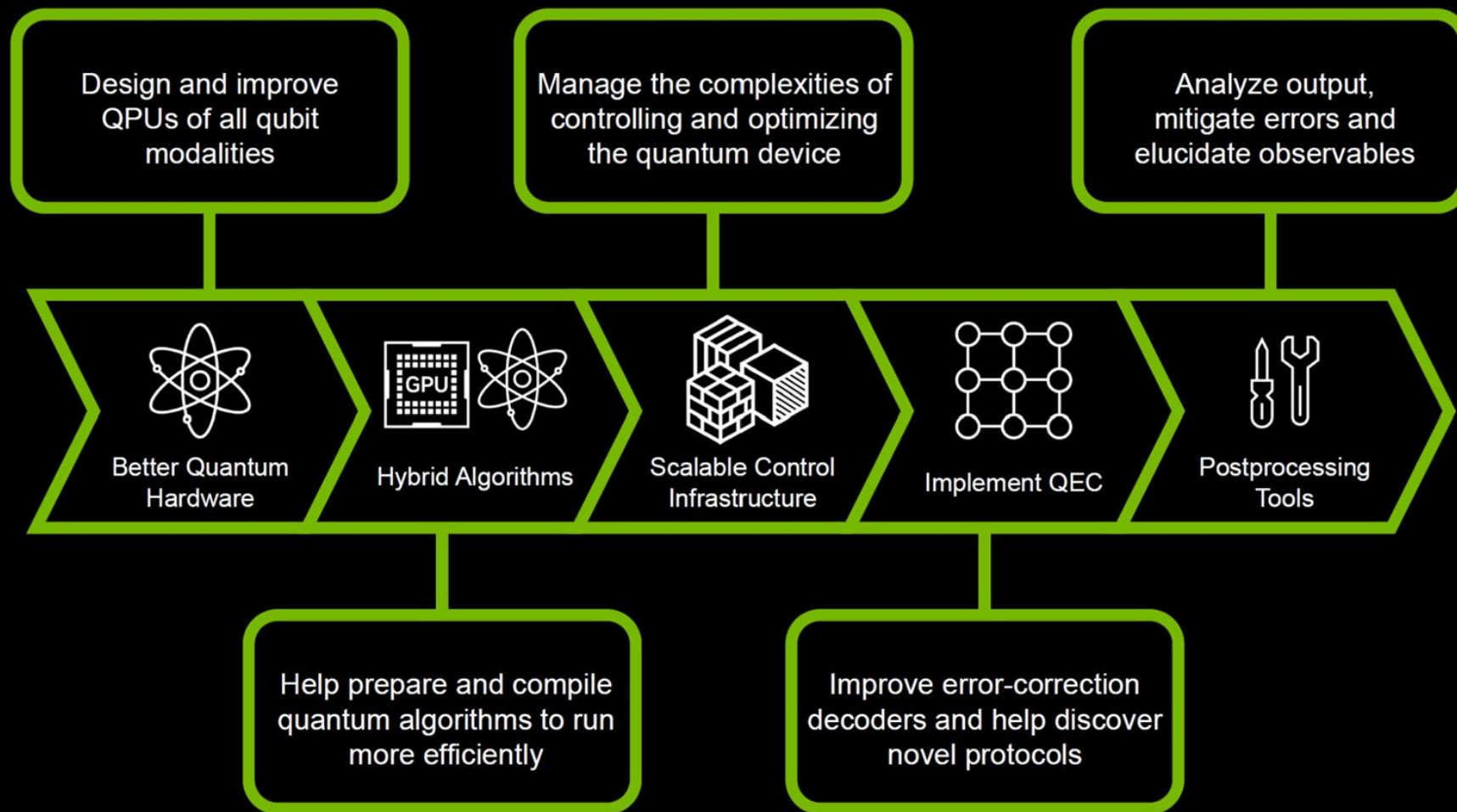
Hybrid Applications

Quantum Data
Generation

Expand Frontier of
Accelerated Computing

Artificial Intelligence for Quantum Computing

[arXiv:2411.09131](https://arxiv.org/abs/2411.09131)



The Generative Quantum Eigensolver

First demonstration of GPT-generated circuits



Challenge

- Variational quantum algorithms offered promise for running drug-discovery applications on small quantum devices - but **suffer from serious optimization issues**.
- Many of these problems are tied to how circuits are parametrized.

Solution

- The generative quantum eigensolver acts like a Large Language Model – but generating quantum circuits from quantum operations, rather than sentences from words.
- Using a generative model like GPT to create quantum circuits **avoids the limitations of traditional variational quantum algorithms**

$\vec{j} \sim$ "Once upon a time . . . happily ever after"
word_{j₁} word_{j₂} word_{j₃} word_{j₄} . . . word_{j_{N-2}} word_{j_{N-1}} word_{j_N} **LLM**

$\vec{j} \sim$
The diagram shows a quantum circuit with multiple horizontal lines representing qubits. The circuit is composed of several layers of gates, each represented by a set of colored squares (black, green, blue, yellow) connected by lines. The layers are labeled U_{j_1} , U_{j_2} , U_{j_3} , and U_{j_N} , with ellipses indicating intermediate layers. The gates are connected by vertical lines, showing the flow of the circuit. **GPT-QE**



New approach in
Using AI for building
quantum applications



Can be
extended to various
application
areas

40X

Speedup over
CPU when
running GQE on
GPU



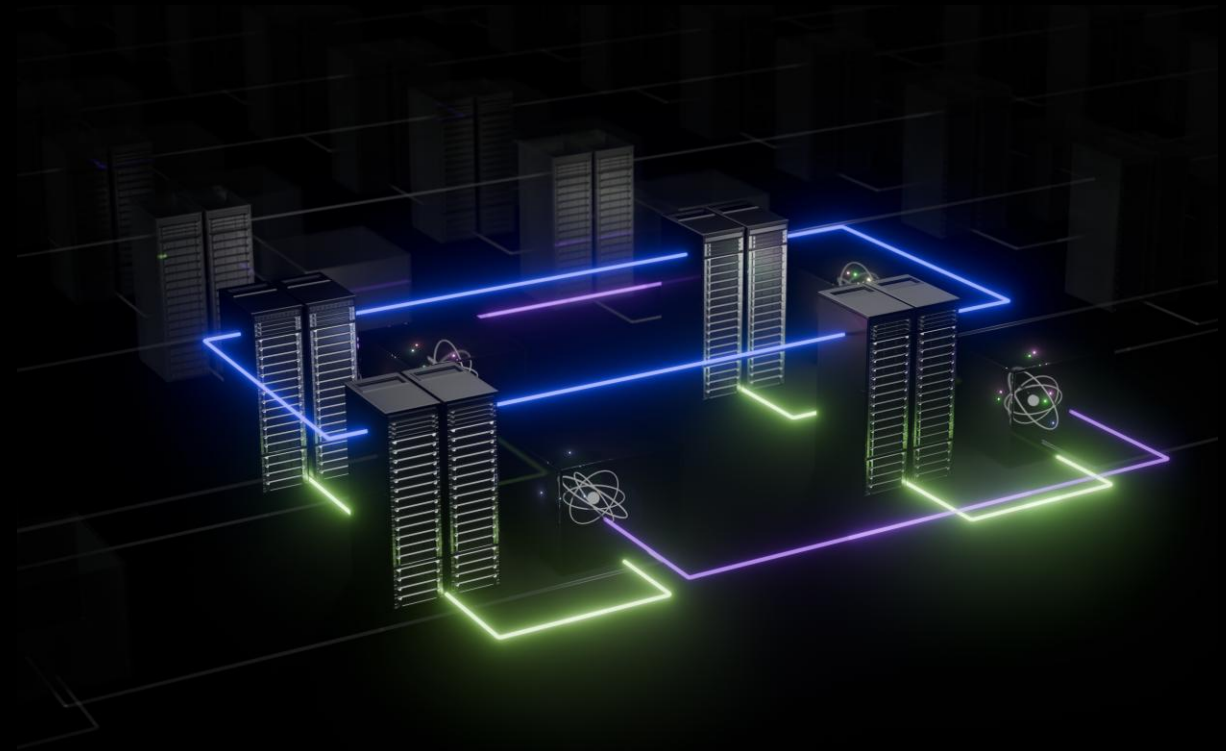
Quantum-GPU Supercomputing

All valuable quantum computing will be **hybrid**

GPUs and AI will prove essential

Petascale compute needed for useful Fault Tolerant Quantum Computers

Developer ecosystem is critical



The NVIDIA Quantum Platform

Bringing AI Supercomputing to Enable Useful Quantum Computing

CUDA-QEC and CUDA-Q Solvers

Turnkey Quantum Research and Application Development

CUDA-Q

Open-Source Software Platform for Accelerated Quantum Supercomputers

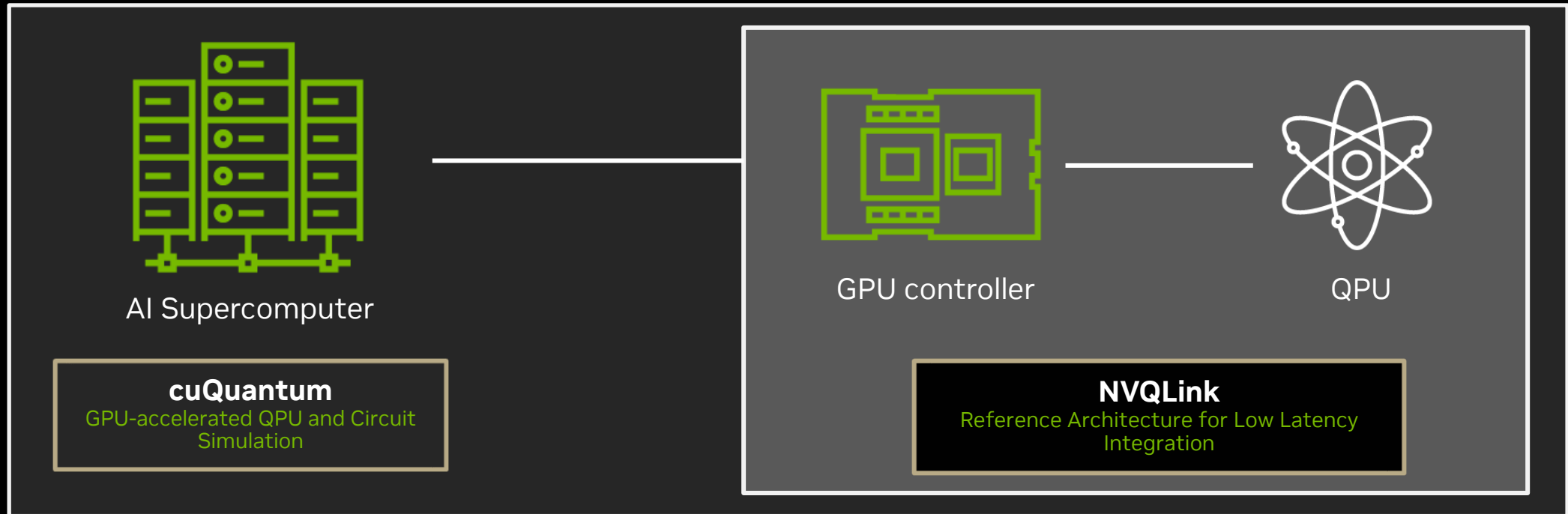
cuQuantum

Accelerated Quantum Simulation/Emulation on GPU

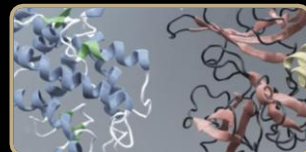
NVQLink

Reference Architecture for Low Latency Integration

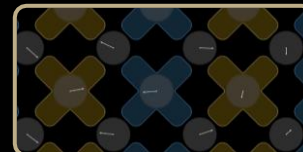
NVIDIA Open Quantum Platform



CUDA-Q
Quantum-GPU Platform

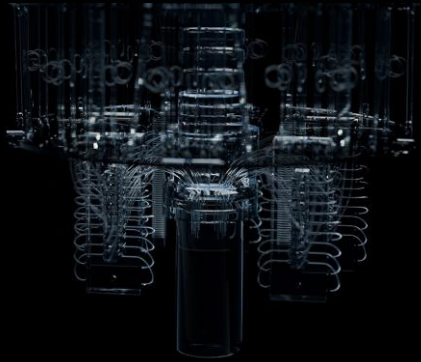


CUDA-Q Solvers

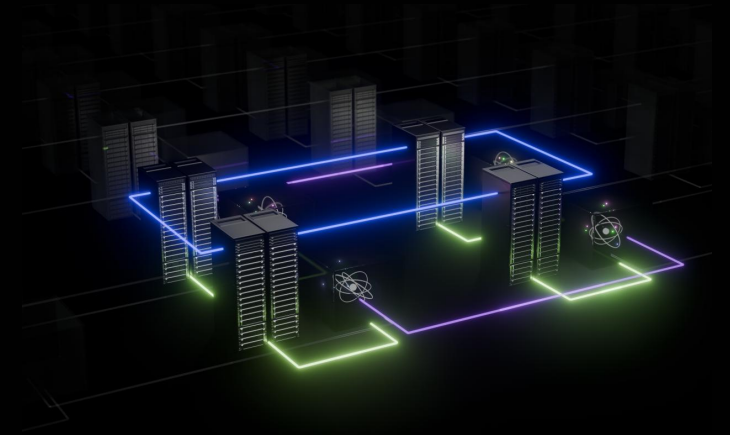


CUDA-Q QEC

The Path from Qubits to Supercomputers

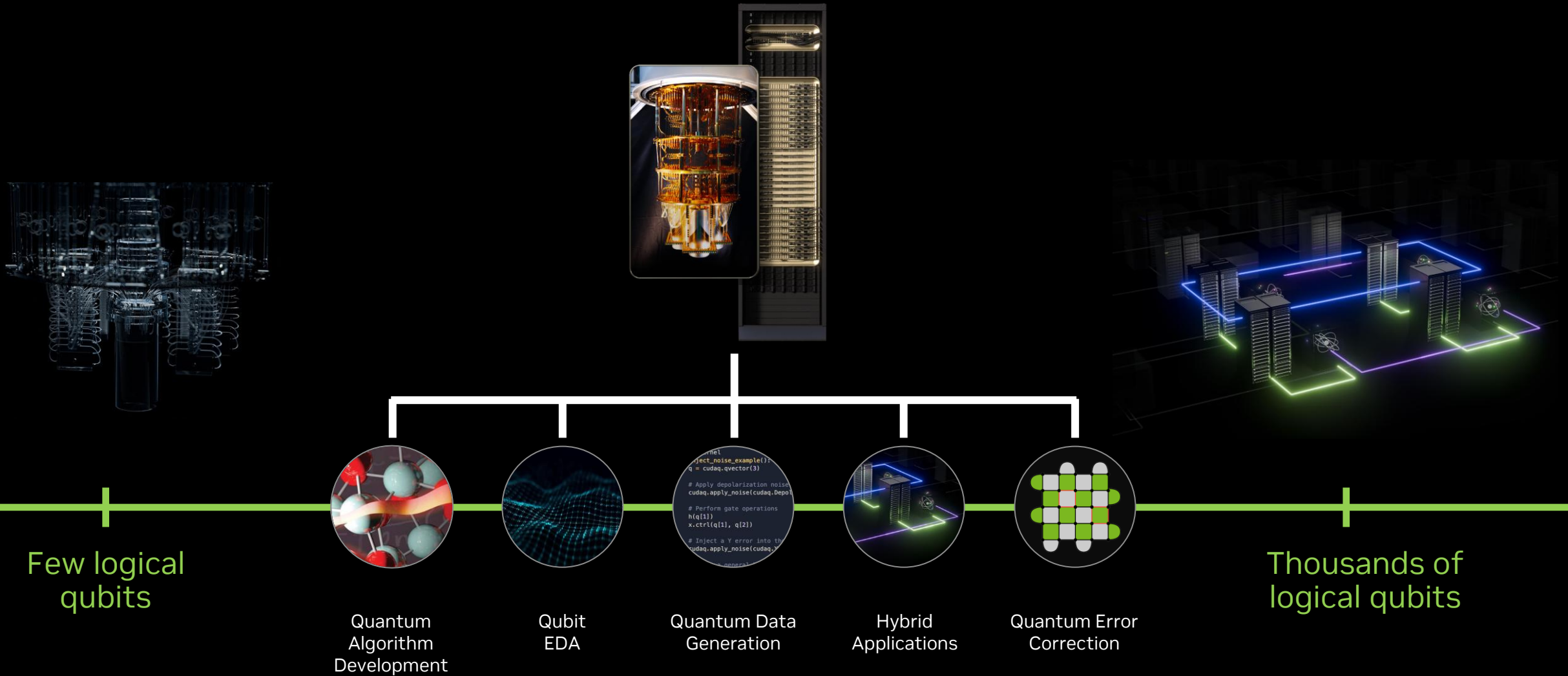


Today's
Systems

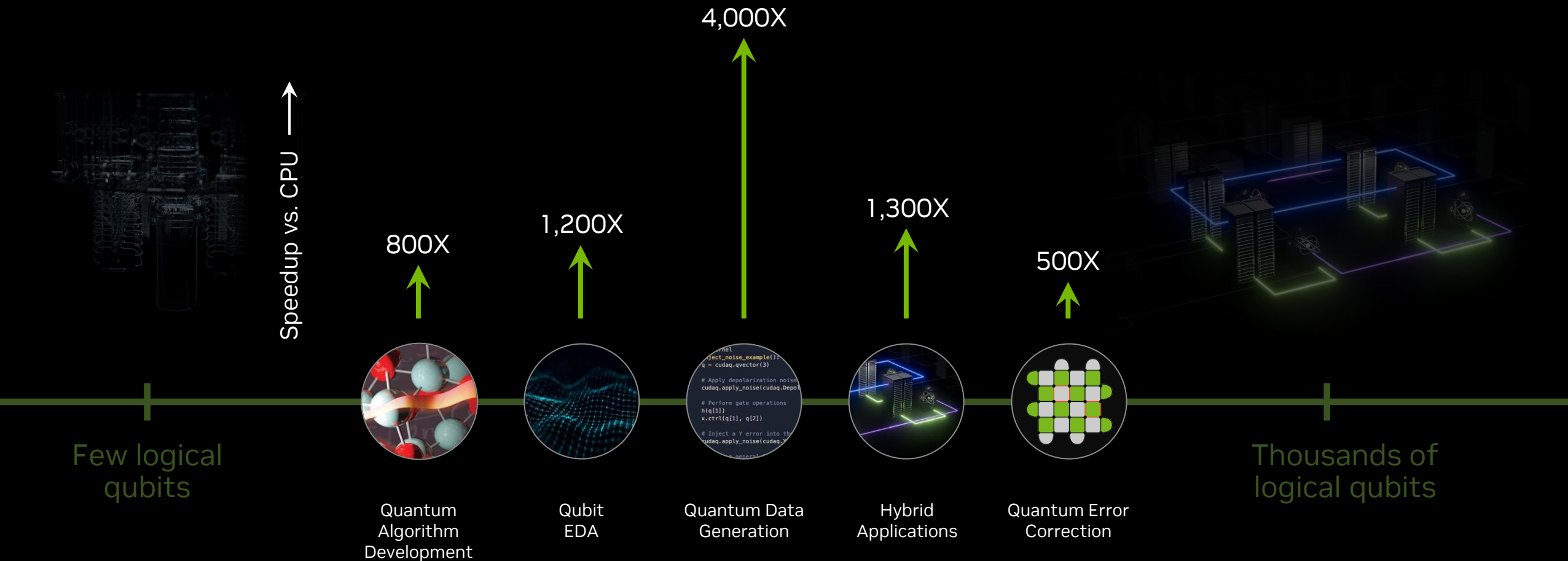


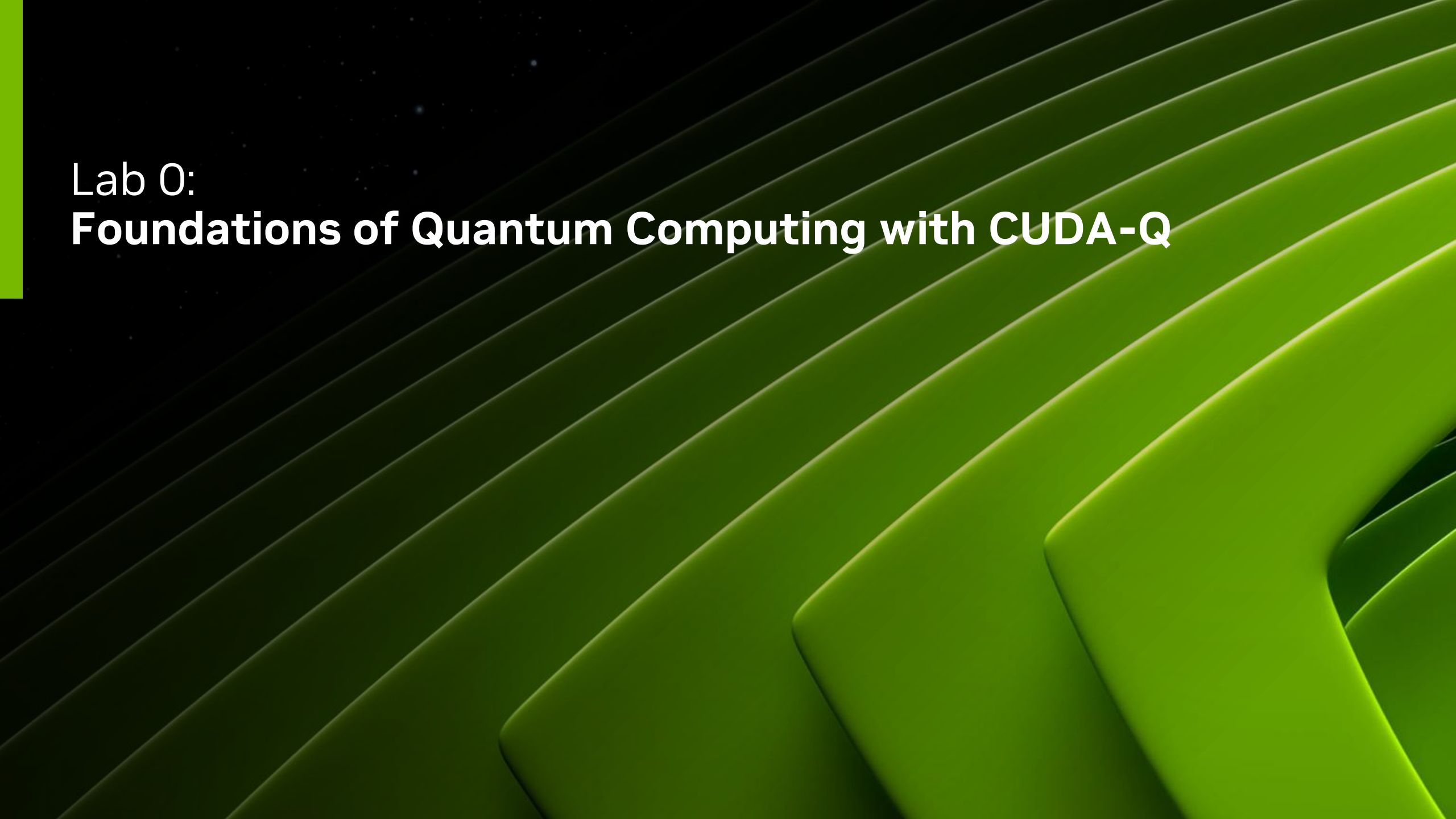
Quantum-GPU
Supercomputers

The Journey to Accelerated Quantum Supercomputing



The Journey to Accelerated Quantum Supercomputing





Lab 0: **Foundations of Quantum Computing with CUDA-Q**